Vidit Agarwal

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EDUCATION

University of California, San Diego — San Diego, CA
Computer Science & Mathematics, Expected June 2028

RELEVANT COURSEWORK

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Cloud Computing, Object-Oriented Programming, Machine Learning

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, C++, SQL
Backend & Infrastructure: FastAPI, React.js, REST APIs, Docker, Kubernetes, AWS, Google Cloud, Linux
Systems & Developer Tools: Git, GitHub Actions, Distributed Systems, Microservices, Cursor, GitHub Copilot, Claude Code, Codex

PROFESSIONAL EXPERIENCE

Software Engineering Intern | *Phrasiq* - *San Francisco, CA* Jul 2026 - Present
Built production backend infrastructure, REST APIs, and full-stack features for an enterprise AI finance platform. Collaborated directly with founding engineers to design, ship, and iterate customer-facing product features.

Full Stack Engineering Intern in Residence | *Circuit Launch* - *Mountain View, CA* Jun 2026 - Present
Developing scalable backend services, system integrations, and cloud-native applications for next-generation physical AI and robotics systems. Building full-stack software infrastructure alongside founding engineers.

Open Source Systems Engineering | *The Linux Foundation* - *Remote* Jan 2026 - Jun 2026
Developed cloud-native backend microservices for an open-source LTE/5G mobile core focused on scalability, reliability, and low-latency communication. Collaborated through Git-based code review, systems debugging, and distributed development workflows.

Blockchain Developer | *Unicage* - *Remote* Jun 2025 - May 2026
Built scalable backend APIs and automation workflows supporting 5K+ decentralized transactions. Designed transaction, payment, and logging infrastructure while integrating distributed backend systems with real-time data pipelines and production SaaS services.

Lead Lab Researcher | *Santa Clara University* - *Santa Clara, CA* Jan 2025 - Jun 2026
Developed full-stack geospatial software using FastAPI, React, Python, Google Earth Engine, and LightGBM to analyze land-cover change and generate environmental insights. Built multimodal AI pipelines combining visual and text-based educational data, improving model performance by 15% through evaluation and optimization.

Data Analyst Intern | *Berkeley Labs* - *Berkeley, CA* Jun 2022 - Aug 2023
Built predictive models using satellite imagery and automated geospatial data pipelines for environmental analysis and vegetation health prediction. Collaborated on remote sensing research involving NDVI, spatial modeling, and environmental prediction.

RELEVANT PROJECTS

Kiro | AI Context Infrastructure for Coding Agents
Built distributed backend infrastructure connecting GitHub repositories, organizational knowledge, and AI coding agents through scalable APIs. Designed retrieval, indexing, and observability workflows to provide agents with consistent project context.

Ido | Universal Agent Harness for 3D Design
Built an AI-powered 3D generation platform converting natural-language prompts into editable CAD models using Blender, OpenSCAD, FastAPI, and agentic code generation. Designed modular prompt-to-IR-to-code pipelines with validation, backend orchestration, and multi-engine adapters.

Wooly | Voice-to-Action Mobile System
Built a real-time voice-controlled system integrating speech recognition, LLM reasoning, structured command generation, and automated mobile actions. Designed backend orchestration for multi-step workflows across messaging, navigation, calls, and application tasks.

ML Labs | Automated ML Research Platform
Built an experimentation platform automating feature engineering, model training, evaluation, and experiment tracking. Developed reproducible testing workflows to compare model iterations, reduce manual experimentation, and identify performance improvements.

Aegis | Adaptive Cyber Defense System with RL Training
Built a reinforcement learning-based system that simulates and responds to cyberattack scenarios. Developed the complete software pipeline for agent training, environment simulation, telemetry processing, and real-time visualization.

AWARDS

Anthropic x AWS Hackathon Winner | Ido | Jun 2026
ML Empowerment Foundation Build Challenge Winner | ML-Labs | Jun 2026
YC x Nozomio Hackathon Track Winner | Kiro | May 2026
AWS Building the Next YC Unicorn Hackathon Winner | ML Labs | Apr 2026
OceanHacks Winner | BlueDocs | Feb 2026
SCU ACM Hackathon Winner | DriveWatch AI | Nov 2024